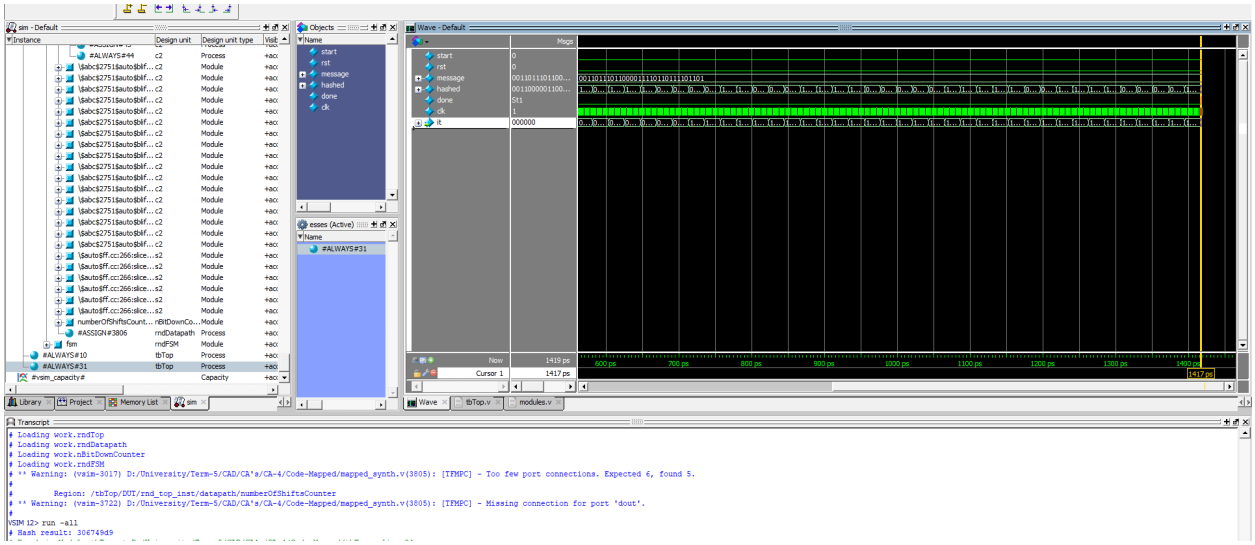
نام و نام خانوادگی: دنیز مصطفی زادگان
شماره دانشجویی: 810102603
نام و نام خانوادگی: سید حمید محمودی
شماره دانشجویی: 810102549
CAD_CA4_REPORT

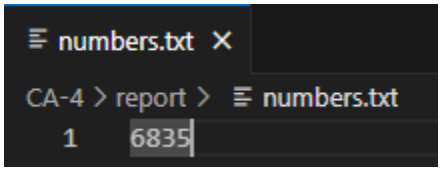
Output-analysis

After using Yosys -s lut2synth.ys to get a synthesized code, we gave it to lut2mapper.py, although we had to change some parts on the .py in order to also map SDFF_PP0P's and SDFF_PP1P's as well to s2 and s1, we also changed the const part back into the normal hardcode form, although we could've left the synthesized code but it took around 200 c2/c1 modules and some s2's, and also we didn't need to have it synthesized.
With that here's the output after running the same tbTop.v on the new code and the python output:

```
Created modules file: modules.v
PS D:\University\Term-5\CAD\CA's\CA-4\Code-mapped> python lut2mapper.py synth.v --cell c2 --count-cells
Found 0 LUT1 and 554 LUT2 instances (out of 554 $lut)
Total number of c2: 554
Total number of S1: 0
Total number of S2: 57
Created mapped design: mapped_synth.v
Created modules file: modules.v
```



And it took 6835 gates which compared to our original 3400 is around 2 times worse but works just fine.



**It is also uploaded as a file with the rest of the report

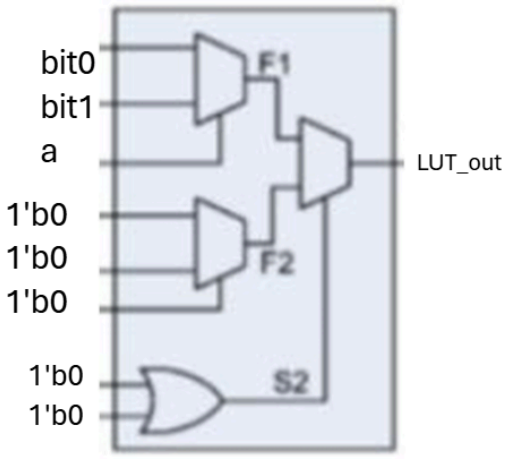
Mapping

First let's take a look at python mapping output:

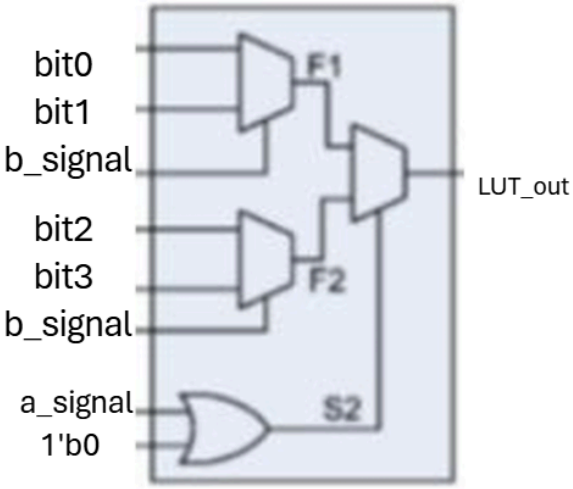
```
Created modules file: modules.v
PS D:\University\Term-5\CAD\CA's\CA-4\Code-mapped> python lut2mapper.py synth.v --cell c2 --count-cells
Found 0 LUT1 and 554 LUT2 instances (out of 554 $lut)
Total number of c2: 554
Total number of S1: 0
Total number of S2: 57
Created mapped design: mapped_synth.v
Created modules file: modules.v
```

Then lets take a look at mapping of lut1 and lut2 to c1 and c2, and dff and sdff to s1 and s2.

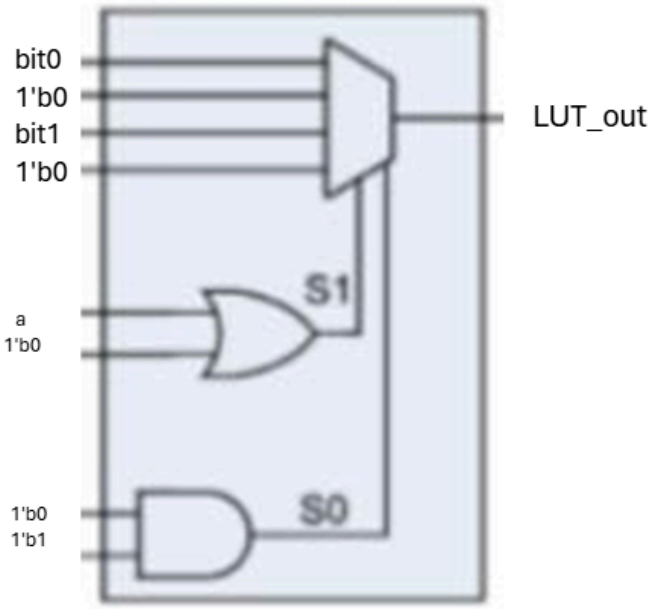
LUT1-C1



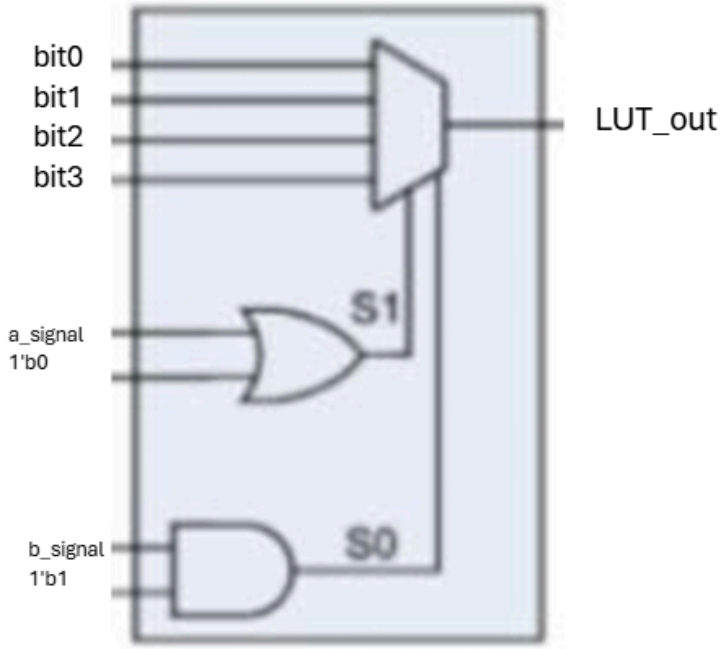
LUT2-C1



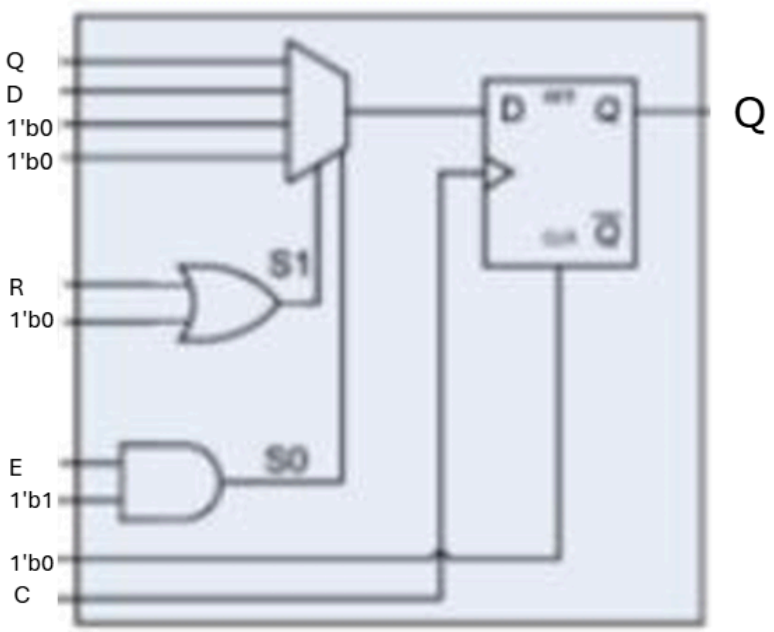
LUT1-C2



LUT2-C2



SDFF-S2



DFF-S1

